Course: Introduction to Artificial Intelligence
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First Choice – LockIn: Personal AI Study Assistant

1. Project Title
LockIn – Personal AI Study Assistant

2. What problem will it solve, and for whom?
University students often struggle with organizing lecture materials, preparing for exams, and managing study time. LockIn will help students study more efficiently by turning their learning materials into organized notes, summaries, and practice quizzes.

3. What will the input and output be? Give one example.

- Input: Lecture PDFs, documents, notes, course names, and quiz answers.
- Output: Structured summaries, generated quizzes, and study recommendations.

Example: A student uploads a lecture about machine learning. LockIn generates a summary and five practice questions to help the student revise.

4. How will AI contribute to the solution?
AI will use Natural Language Processing (NLP) to summarize lectures, generate questions, and identify important concepts. Later, it could analyze quiz results and recommend topics that need more revision.

5. What data or resources will you need?
Student-provided lecture materials, Python, PDF text-extraction tools, an NLP model or API, and a simple web interface. Sample lectures will be used for testing.

6. What small, working version could you realistically build this semester?
A web application where students upload a lecture PDF, receive an AI-generated summary, and complete a short quiz based on the material.

7. How will you test whether it works?
I will test the application with different lecture documents, compare summaries with the original content, and check whether generated quiz questions are accurate and relevant.

8. What is the main challenge or risk?
The main challenge is ensuring AI-generated summaries and questions are correct. Incorrect information could mislead students, so outputs must be checked against the original materials.

Second Choice – MapOut: Complete AI Travel Planner

1. Project Title
MapOut – Complete AI Travel Planner

2. What problem will it solve, and for whom?
Planning a trip requires researching attractions, transportation, accommodation, and costs across multiple platforms. MapOut will help travelers organize personalized trips according to their budget, interests, and available time.

3. What will the input and output be? Give one example.

- Input: Destination, travel dates, budget, number of travelers, and interests.
- Output: Personalized daily itinerary, recommended attractions, and estimated trip costs.

Example: A student plans a three-day trip to Rome with a €300 budget. MapOut recommends historical attractions, organizes daily activities, and estimates costs.

4. How will AI contribute to the solution?
AI will use recommendation algorithms to select suitable attractions and planning algorithms to organize activities. Optimization techniques will help group nearby locations and keep the itinerary within time and budget limits.

5. What data or resources will you need?
Attraction and location data from sources such as OpenStreetMap, estimated prices, Python, and a simple web interface. Weather information could be added later.

6. What small, working version could you realistically build this semester?
A basic travel planner for one city where users enter their budget, interests, and trip duration. The application generates a daily itinerary with recommended attractions and estimated activity costs.

7. How will you test whether it works?
I will test different travel preferences, budgets, and trip durations to check whether the recommendations are relevant, the itinerary is realistic, and estimated costs stay within the selected budget.

8. What is the main challenge or risk?
The main challenge is combining different travel requirements into one realistic itinerary. Prices, opening hours, and availability may change, making accurate planning difficult.

Third Choice – CookUp: Personalized Meal Planning Assistant

1. Project Title
CookUp – Personalized Meal Planning Assistant

2. What problem will it solve, and for whom?
Students and busy individuals often struggle to plan affordable meals using available ingredients. CookUp will help users save time, manage food expenses, and reduce waste through personalized meal recommendations.

3. What will the input and output be? Give one example.

- Input: Budget, available ingredients, food preferences, dietary restrictions, and cooking time.
- Output: Recommended recipes, a weekly meal plan, shopping list, and estimated costs.

Example: A student enters a €40 weekly budget and ingredients such as rice, chicken, and eggs. CookUp recommends suitable meals and creates a shopping list.

4. How will AI contribute to the solution?
AI will use recommendation algorithms to match recipes with user preferences and available ingredients. Constraint-satisfaction and optimization techniques will help create meal plans that respect budgets, dietary restrictions, and cooking time.

5. What data or resources will you need?
A recipe dataset containing ingredients and preparation times, estimated ingredient prices, Python, and a simple web interface.

6. What small, working version could you realistically build this semester?
An application with a small recipe database where users enter their preferences and receive recommended meals, a basic weekly plan, and an estimated shopping list.

7. How will you test whether it works?
I will test different budgets, ingredients, and dietary restrictions to check whether the recommendations are suitable, restricted ingredients are excluded, and estimated costs are calculated correctly.

8. What is the main challenge or risk?
The main challenge is balancing multiple requirements while keeping recommendations accurate. Ingredient prices may change, and dietary restrictions must always be respected.

Why is my first choice the best option for me?

My first choice is **LockIn** because it solves a problem I personally understand as a university student. Organizing lecture materials and preparing for exams can be time-consuming, and I believe an AI study assistant could make this process easier.

It is also the most realistic project to develop within one semester because I can start with basic features such as PDF uploading, summarization, and quiz generation, then gradually improve the application.

Compared with **CookUp** and **MapOut**, **LockIn** requires less external data and is easier to test using real lecture materials. It also gives me the opportunity to explore NLP and AI-based personalization while creating something useful for students.

What will I complete before the next class?

Before the next class, I plan to:

1. Define the main features and technical requirements of LockIn.
2. Collect sample lecture PDFs for development and testing.
3. Select the tools for PDF extraction and AI summarization.
4. Design a simple interface for uploading and viewing lecture materials.
5. Begin a basic prototype that extracts text and generates a summary.

My goal is to demonstrate the initial document-processing workflow and explain the next development steps.

AI Use Disclosure

I developed my three project ideas independently and used ChatGPT (OpenAI) to refine and improve them. I asked for suggestions on possible features, AI techniques, and how to make each project realistic for development within one semester. I reviewed the suggestions, selected what was useful, and made my own decisions about the final concepts and their scope.

All three ideas are my own, and AI was used only for clarification, improvement, and organization. At the end, I checked that each idea met the professor’s requirements and could realistically be developed and tested.